Bias Reduction & Fairness Strategy

The system incorporates fairness mechanisms to ensure unbiased candidate evaluation:

1. **Text Normalization** All resumes and job descriptions are converted into a standardized format to eliminate formatting inconsistencies.
2. **Personal Information Masking** Sensitive attributes such as names, email addresses, and phone numbers are removed to prevent bias.
3. **Balanced Scoring** Keyword-based scoring is balanced with semantic similarity to avoid over-reliance on keyword matching.
4. **Score Normalization** All scores are normalized to maintain consistency across different candidates.
5. **Bias Detection** The system flags potential bias indicators such as gender, age, or marital status references.